Okan Bulut

Professor, Measurement, Evaluation, and Data Science
Faculty of Education, University of Alberta

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EDUCATION

University of Minnesota, Minneapolis, MN, USA

Ph.D., Educational Psychology - Quantitative Methods in Education, June 2013

M.A., Educational Psychology - Quantitative Methods in Education, January 2010

Adnan Menderes University, Aydin, TURKEY

B.A., Elementary Education, June 2006

ACADEMIC POSITIONS

Professor July, 2024 to Now
University of Alberta (Edmonton, AB)
Faculty of Education
Program: Measurement, Evaluation, and Data Science (MEDS)

Associate Professor July, 2018 to June, 2024
University of Alberta (Edmonton, AB)
Faculty of Education
Program: Measurement, Evaluation, and Data Science (MEDS)

Assistant Professor September, 2014 to June, 2018
University of Alberta (Edmonton, AB)
Faculty of Education
Program: Measurement, Evaluation, and Data Science (MEDS)

NON-ACADEMIC POSITIONS

Psychometrician May, 2013 to October, 2014
American Institutes for Research (Washington, DC)
Assessment Program
Project: Statewide Alternate Assessments for Students with Disabilities

Research Fellow September, 2012 to May, 2013
Pearson Assessments (Bloomington, MN)
Clinical Assessments
Project: Technical Evaluation of Behavioral and Intelligence Tests

RESEARCH INTERESTS

Artificial intelligence in education, educational data mining, learning analytics, computerized and digital assessments, and computational psychometrics

TEACHING EXPERIENCE

- Instructor and course developer, *Analyzing Survey Data with R* (Pluralsight)
- Instructor and course developer, *Reshaping Data with R* (Pluralsight)
- Instructor and course developer, *EDPY 607 Measurement Theory II* (U of Alberta)
- Instructor and course developer, *EDPY 504 Survey Design & Implementation* (U of Alberta)
- Instructor, *EDPY 303 Educational Assessment* (U of Alberta)
- Instructor, *EDPY 501 Introduction to Data Analysis in Educational Research* (U of Alberta)
- Instructor, *EDPY 506 Machine Learning* (U of Alberta)
- Instructor, *EDPY 507 Measurement Theory I* (U of Alberta)
- Instructor, *EDPY 605 Quantitative Methods II* (U of Alberta)

CURRENT RESEARCH GRANTS (TOTAL: $587,541)

Identifying Adaptive Problem-Solving Profiles to Mitigate Systemic Barriers in Canada's Labour Market

- **Role:** Principal Investigator
- **Grant Title:** Strengthening Canada's Skills with PIAAC Research
- **Funding Source:** Future Skills Centre
- **Awarded Amount:** $88,480 (2026)

Evaluating the Impact of Generative AI on the Sustainability of Assessments in Higher Education

- **Role:** Principal Investigator
- **Grant Title:** Research Development Fund
- **Funding Source:** Worldwide Universities Network & University of Alberta
- **Awarded Amount:** £9,628 + $5000 (2025-2026)

Does Explainable AI Shape Trust and Critical AI Adoption Across Continents?

- **Role:** Co-PI (PI: Nino Gugushvili - Maastricht University)
- **Grant Title:** Research Development Fund
- **Funding Source:** Worldwide Universities Network
- **Awarded Amount:** £9,969 (2026-2027)

Improving Canadian Teachers' Assessment Literacy and Decision-Making through Self-Directed Professional Development

- **Role:** Principal Investigator
- **Grant Title:** Insight Grant
- **Funding Source:** Social Sciences and Humanities Research Council of Canada
- **Awarded Amount:** $241,984 (2023-2028)

Enhancing Students' Reasoning and Decision-making Skills: Science, Values, and Contradictions in Socioscientific Contexts

- **Role:** Co-PI (PI: Mijung Kim - University of Alberta)
- **Grant Title:** Insight Grant
- **Funding Source:** Social Sciences and Humanities Research Council of Canada
- **Awarded Amount:** $140,379 (2023-2028)

Many Teacher Emotions: Setting the Stage for a Global Study on Teachers' Emotions

- **Role:** Co-PI (PI: Lia Daniels - University of Alberta)
- **Grant Title:** KIAS Cluster Grant
- **Funding Source:** The Kule Institute for Advanced Study
- **Awarded Amount:** $75,000 (2025-2027)

PREVIOUS RESEARCH GRANTS

Identifying the Predictors of Mathematics Anxiety, Confidence, and Performance in Canada: An Educational Data Mining Approach

- **Role:** Principal Investigator
- **Grant Title:** Research Partnership Grant
- **Funding Source:** Alberta Education
- **Awarded Amount:** $14,960 (2022-2024)

Examining Scoring Variability of the Casper Test: An Application of Explanatory Item Response Modeling

- **Role:** Principal Investigator
- **Grant Title:** Alo Grant
- **Funding Source:** Altus Assessments Inc.
- **Awarded Amount:** $20,000 USD (2021-2024)

Factors Influencing Observers' Interpretations of School Principals' Gestures

- **Role:** Co-PI (PI: Darryl Hunter - University of Alberta)
- **Grant Title:** Killam Research Fund Research Cornerstone Grant
- **Funding Source:** University of Alberta
- **Awarded Amount:** $48,000 (2023-2024)

Climate Change Surveillance for Chronic Health Effects in Vulnerable Populations

- **Role:** Co-Investigator (PI: Shelby Yamamoto - University of Alberta)
- **Grant Title:** Enhanced Surveillance for Chronic Disease Program
- **Funding Source:** Public Health Agency of Canada
- **Awarded Amount:** $962,322 (2020-2024)

Preparing Cities for Climate Change

- **Role:** Co-Investigator (PI: Shelby Yamamoto - University of Alberta)
- **Grant Title:** Operating Grants: Data Analysis Using Existing Databases and Cohorts
- **Funding Source:** Canadian Institutes of Health Research
- **Awarded Amount:** $98,271 (2022-2023)

Providing Students with a Voice: A Measure of Students' Socio-Emotional Well-Being to Optimize Learning for All Students

- **Role:** Principal Investigator
- **Grant Title:** Research Partnership Grant
- **Funding Source:** Alberta Education
- **Awarded Amount:** $45,123 (2019-2022)

Automatic Generation of Passages for Reading Comprehension: An Artificial Intelligence Approach

- **Role:** Principal Investigator
- **Grant Title:** Support for the Advancement of Scholarship (SAS) Grant
- **Funding Source:** University of Alberta
- **Awarded Amount:** $6,895 (2020-2021)

Integrating Theory and Practice to Enhance Inclusive Education in an Undergraduate Teacher Program

- **Role:** Co-Investigator (PI: Denyse Hayward - University of Alberta)
- **Grant Title:** Alberta Education Conditional Grant Funding
- **Funding Source:** Alberta Education
- **Awarded Amount:** $49,276 (2018-2022)

Public Health Messaging Insights in Alberta During Wave 1 of COVID-19: A Complexity-Informed Mixed Methods Design for Rapid Response to Big Data Integration

- **Role:** Co-Investigator (PI: Cheryl Poth - University of Alberta)
- **Grant Title:** Support for the Advancement of Scholarship (SAS) Grant
- **Funding Source:** University of Alberta
- **Awarded Amount:** $7,000 (2020-2021)

Climate Change, Older Adults and Immigrants: Exploring Community Vulnerability and Resilience

- **Role:** Co-Investigator (PI: Shelby Yamamoto - University of Alberta)
- **Grant Title:** Cities IPCC Legacy Research Grant Program
- **Funding Source:** The City of Edmonton & Alberta Ecotrust
- **Awarded Amount:** $50,000 (2019-2021)

Improving the Quality and Quantity of Feedback via Interactive Score Reporting

- **Role:** Principal Investigator
- **Grant Title:** Teaching and Learning Enhancement Fund (TLEF)
- **Funding Source:** University of Alberta
- **Awarded Amount:** $62,521 (2018-2021)

Building Community Climate Change Knowledge and Resilience in Older Adults and Immigrants

- **Role:** Co-Investigator (PI: Shelby Yamamoto - University of Alberta)
- **Grant Title:** North American Partnership for Environmental Community Action (NAPECA)
- **Funding Source:** Commission for Environmental Cooperation
- **Awarded Amount:** $155,000 (2019-2021)

Developing an Interactive Score Reporting System to Improve the Quality of Formative Feedback in Higher Education

- **Role:** Principal Investigator
- **Grant Title:** Insight Development Grant
- **Funding Source:** Social Sciences and Humanities Research Council of Canada
- **Awarded Amount:** $59,994 (2016-2019)

Effect of Detailed Score Reporting on Student Learning

- **Role:** Co-Investigator (PI: Vijay Daniels - University of Alberta)
- **Grant Title:** Teaching and Learning Enhancement Fund (TLEF)
- **Funding Source:** University of Alberta
- **Awarded Amount:** $59,171 (2017-2019)

Examining the Measurement Properties of What Albertans Know about Fetal Alcohol Spectrum Disorders Survey

- **Role:** Principal Investigator
- **Grant Title:** SAGE Secondary Data Use Grant
- **Funding Source:** PolicyWise for Children & Families
- **Awarded Amount:** $5,000 (2018-2019)

Trajectories in Student Mathematics Performance

- **Role:** Research Partner (PI: Lynn McGarvey - University of Alberta)
- **Grant Title:** The Research Partnership Program
- **Funding Source:** Alberta Education
- **Awarded Amount:** $24,100 (2017-2018)

Improving Instructors' Understanding of Universal Student Ratings of Instruction (USRI) at the Faculty of Education: A Pilot Study

- **Role:** Principal Investigator
- **Grant Title:** Support for the Advancement of Scholarship (SAS) Grant
- **Funding Source:** University of Alberta
- **Awarded Amount:** $6,807 (2017-2018)

Understanding Nonresponse Behaviours of Special Education Students in Reading and Mathematics Assessments Using Cognitive Diagnostic Modelling

- **Role:** Principal Investigator
- **Grant Title:** Support for the Advancement of Scholarship (SAS) Grant
- **Funding Source:** University of Alberta
- **Awarded Amount:** $6,384 (2015-2016)

TRAVEL & PROFESSIONAL DEVELOPMENT GRANTS

Teaching and Learning Enhancement Fund

- **Funding Source:** University of Alberta
- **Awarded amount:** $2,860 (2018-2019)

Killam Research Fund – Cornerstones Grant Program

- **Funding Source:** University of Alberta
- **Awarded amount:** $4,200 (2017-2018)

Support for the Advancement of Scholarship (SAS) Travel Grant

- **Funding Source:** University of Alberta
- **Awarded amount:** $1,200 (2015-2016)

AWARDS

- Martha Cook Piper Research Prize by the University of Alberta 2023 to 2024
- Journal of Intelligence Young Investigator Award 2022 to 2023
- Coutts-Clarke Fellowship by University of Alberta 2021 to 2022
- Distinguished Paper Award by AERA 2017 to 2018
- Alicia Cascallar Outstanding Paper Award by NCME 2015 to 2016
- John P. Yackel/Pearson Assessment Graduate Fellowship 2012 to 2013
- Research Grant from Applied Psychological Measurement, Inc. 2012 to 2013
- Mary Corcoran Graduate Fellowship at the University of Minnesota 2011 to 2012
- Graduate Scholarship from the Turkish Ministry of National Education 2007 to 2009

BOOKS

1. **Bulut, O.**, Shin, J., Yildirim-Erbasli, S., & Desjardins, C. D. (In preparation). *Educational data mining: Applications in digital assessments*. Boca Raton, FL: CRC Press.
2. **Bulut, O.** (2020). *Conducting Monte Carlo simulations in R*. University of Alberta Education and Research Archive. doi:10.7939/r3-y066-fs55
3. **Bulut, O.** (2024). *Psycho-educational assessments: Theory and practice*. MDPI. doi:10.3390/books978-3-7258-0551-8
4. Desjardins, C. D., & **Bulut, O.** (2018). *Handbook of educational measurement and psychometrics using R*. Boca Raton, FL: CRC Press.

BOOK CHAPTERS

1. **Bulut, O.**, & Walsh, C. (2026). Closing the feedback loop: Automated quality assurance for LLM-generated feedback. In E. G. Blanchard, G. Chen, M. Chi, & S. Isotani (Eds.), *Artificial Intelligence in Education. Late Breaking Results, WideAIED, Practitioners, Industry and Policies, Blue Sky, Doctoral Consortium, FoL Workshops and Tutorials, FoL Invited Papers* (Communications in Computer and Information Science, Vol. 3031, pp. 130–136). Springer. doi:10.1007/978-3-032-29788-4_19
2. **Bulut, O.**, Tan, B., & Yildirim-Erbasli, S. N. (2026). Does conversation-based assessment elicit higher-level thinking? Evidence from expert and LLM ratings. In E. G. Blanchard, G. Chen, M. Chi, & S. Isotani (Eds.), *Artificial Intelligence in Education. Late Breaking Results, WideAIED, Practitioners, Industry and Policies, Blue Sky, Doctoral Consortium, FoL Workshops and Tutorials, FoL Invited Papers* (Communications in Computer and Information Science, Vol. 3031, pp. 222–229). Springer. doi:10.1007/978-3-032-29788-4_32
3. Yuan, Z., **Bulut, O.**, Geoffre, T., & Wang, Q. (2026). From assessment to human–AI co-creation in language learning: Adaptive, inclusive, and game-based design in the generative AI era. In E. G. Blanchard, G. Chen, M. Chi, & S. Isotani (Eds.), *Artificial Intelligence in Education. Late Breaking Results, WideAIED, Practitioners, Industry and Policies, Blue Sky, Doctoral Consortium, FoL Workshops and Tutorials, FoL Invited Papers* (Communications in Computer and Information Science, Vol. 3033, pp. 39–43). Springer. doi:10.1007/978-3-032-29794-5_7
4. Jerez, D., & **Bulut, O.** (2026). Tracing the paths of slow responders in complex problem-solving: A transition network analysis approach. In M. Saqr, K. Misiejuk, & S. López-Pernas (Eds.), *Innovations in analytics of learning dynamics* (pp. 114–124). Springer Nature Switzerland. doi:10.1007/978-3-032-34157-0_9
5. **Bulut, O.**, Liu, J. X., & Liu, X. (2026). Leveraging AI-powered tools for personalized learning pathways. In S. Papadakis (Ed.), *Virtual tutors and AI-powered instructional tools in K-12 settings* (pp. 173-220). IGI Global Scientific Publishing. doi:10.4018/979-8-3373-2637-5.ch006
6. **Bulut, O.**, Tan, B., & Mazzullo, E. (2026). Applications of large language models in automatic item and feedback generation for early elementary education. In S. Papadakis (Ed.), *AI applications in preschool and primary education – Teaching with artificial intelligence* (pp. 209–225). Springer. doi:10.1007/978-3-032-17190-0_12
7. Gorgun, G., **Bulut, O.**, & Tan, B. (2025). Designing classroom assessments with generative AI: A teacher-in-the-loop framework. In S. Papadakis (Ed.), *Teaching with artificial intelligence: A guide for primary and elementary educators* (pp. 162-180). Routledge. doi:10.4324/9781003685241-14
8. **Bulut, O.**, Liu, J. X., & Demir, H. (2025). Deep reinforcement learning for engagement-aware question selection in adaptive assessment systems. In A. I. Cristea, E. Walker, Y. Lu, O. C. Santos, & S. Isotani (Eds.), *Artificial intelligence in education. Posters and late breaking results, workshops and tutorials, industry and innovation tracks, practitioners, doctoral consortium, blue sky, and wideAIED* (pp. 27-34). AIED 2025. Communications in Computer and Information Science, vol 2592. Springer, Cham. doi:10.1007/978-3-031-99267-4_4
9. Wongvorachan, T., & **Bulut, O.** (2025). The use of natural language processing in learning analytics. In M. Saqr & S. López-Pernas (Eds.), *Advanced learning analytics methods. AI, Precision and Complexity* (pp. 207-234). Springer, Cham. doi:10.1007/978-3-031-95365-1_9
10. Wongvorachan, T., & **Bulut, O.** (2025). The advanced applications of psychological networks with exploratory graph analysis. In M. Saqr & S. López-Pernas (Eds.), *Advanced learning analytics methods. AI, Precision and Complexity* (pp. 313-352). Springer, Cham. doi:10.1007/978-3-031-95365-1_13
11. **Bulut, O.** (2025). Creating value from process data: Implications for multistage testing. In D. Yan, A. von Davier, & D. Weiss (Eds.), *Research for practical issues and solutions in computerized multistage testing* (pp. 242-261). Routledge. doi:10.4324/9780429263491-16
12. **Bulut, O.** (2024). From adaptive testing to personalized adaptive testing: Applications of recommender systems. In H. Jiao & R. W. Lissitz (Eds.), *Machine learning, natural language processing, and psychometrics*

(pp. 177-199). Information Age Publishing.

13 **Bulut, O.**, Yildirim-Erbasli, S. N., & Gorgun, G. (2024). Assessment analytics for digital assessments: Identifying, modeling, and interpreting behavioral engagement. In M. Sahin & D. Ifenthaler (Eds.), *Assessment analytics in education – Designs, methods and solutions* (pp. 35-60). Springer, Cham. doi:10.1007/978-3-031-56365-2_3

14 Yildirim-Erbasli, S. N., Gorgun, G., & **Bulut, O.** (2024). Enhancing self-regulated learning with artificial intelligence-powered learning analytics. In N. K. Ulutaş & D. Höl (Eds.), *Emergent practices of learning analytics in K-12 classrooms* (pp. 57-83). IGI Global. doi:10.4018/979-8-3693-0066-4.ch004

15 **Bulut, O.**, & Cui, Y. (2021). Validating the findings of a survey research study. In U. Luhanga & A. G. Harbaugh (Eds.), *Basic elements of survey research in education: Addressing the problems your advisor never told you about* (pp. 719-746). Charlotte, NC: Information Age Publishing.

16 Gierl, M. J., **Bulut, O.**, & Zhang, X. (2018). Using computerized formative testing to support personalized learning in higher education: An application of two assessment technologies. In R. Zheng (Ed.), *Digital technologies and instructional design for personalized learning* (pp. 99-119). Hershey, PA: IGI Global. doi:10.4018/978-1-5225-3940-7.ch005

NON-REFEREED ARTICLES

1 **Bulut, O.** (2021). Lexicon-based sentiment analysis using R. Towards Data Science. https://towardsdatascience.com/.

2 **Bulut, O.** (2021). Likert scales: Friend or foe? Towards Data Science. https://towardsdatascience.com/.

3 **Bulut, O.** (2021). AI can bring fairness to assessments but are we ready for it? Towards AI. https://medium.com/towards-artificial-intelligence.

4 **Bulut, O.** (2021). How AI will end the one-size-fits-all approach in human assessment. Towards AI. https://medium.com/towards-artificial-intelligence.

5 **Bulut, O.** (2021). Too many survey questions? Let ant colony optimization handle it. Towards Data Science. https://towardsdatascience.com/.

6 **Bulut, O.** (2021). Effective feature selection: Recursive feature elimination using R. Towards Data Science. https://towardsdatascience.com/.

7 **Bulut, O.** (2020). Testing for measurement invariance in R. Towards Data Science. https://towardsdatascience.com/.

REFEREED JOURNAL ARTICLES & PREPRINTS

1 **Bulut, O.** (In press). Auditing the chain: Algorithmic opacity, criterion bias, and fairness drift in AI-supported measurement. A commentary on Cheng. *Psychometrika*.

2 **Bulut, O.**, & Daniels, L. M. (In press). From black boxes to open books: Leveraging explainable AI to sustain student motivation. *Educational Studies*.

3 Jerez, D., **Bulut, O.**, & Johnson, M. (2026). From orientation to action: How thinking longer at the start matters for complex problem-solving efficiency. *Intelligence, 119,* Article 102074. doi:10.1016/j.intell.2026.102074

4 Yildirim-Erbasli, S. N., Ilgun Dibek, M., & **Bulut, O.** (2026). Same meaning, different words: Automated item paraphrasing in ESP reading assessment. *English for Specific Purposes, 85,* 71-81. doi:10.1016/j.esp.2026.09.003

5 Deribo, T., **Bulut, O.**, & Arslan, B. (2026). Motivational test instructions and conditional structure of cognitive disengagement in low-stakes assessments. *PsychArchives*. doi:10.23668/psycharchives.22563

6 Bello, B., **Bulut, O.**, Hindmarch, W., & McCabe, E. (2026). Measurement invariance of the Pediatric Quality of Life Index in a child and adolescent mental health population. *Quality of Life Research, 34*,

Article 300. doi:10.1007/s11136-026-04403-9

7 Mazzullo, E., & **Bulut, O.** (2026). Diversity and carelessness in LLM-generated psychometric data: Implications for data quality and trust. *Computers, 15*(9), Article 562. doi:10.3390/computers15090562

8 Connolly, D., Salvalaggio, G., **Bulut, O.**, Cottrell-McDermott, C., Hyshka, E., Taylor, M., Piggott, B., Staines, A., Issa, T., Collins, Z., Gehring, N. D., Twan, S., & Morris, H. (2026). Factors associated with willingness of people who smoke unregulated drugs to use supervised inhalation sites. *International Journal of Drug Policy, 10*, Article 105445. doi:10.1016/j.drugpo.2026.105445

9 **Bulut, O.** (2026). Review of the Classic Learning Test 3-8. In J. F. Carlson, K. F. Geisinger, J. L. Jonson, & N. A. Anderson (Eds.), *The twenty-second mental measurements yearbook* (pp. 167-169). Lincoln, NE: Buros Center for Testing.

10 **Bulut, O.** (2026). Review of the Comprehensive Mathematical Abilities Test-Second Edition. In J. F. Carlson, K. F. Geisinger, J. L. Jonson, & N. A. Anderson (Eds.), *The twenty-second mental measurements yearbook* (pp. 188-190). Lincoln, NE: Buros Center for Testing.

11 **Bulut, O.** (2026). Review of the MAP Growth. In J. F. Carlson, K. F. Geisinger, J. L. Jonson, & N. A. Anderson (Eds.), *The twenty-second mental measurements yearbook* (pp. 438-440). Lincoln, NE: Buros Center for Testing.

12 Yaylaci, F. T., Özbey-Demir, Ö., Akbaş, B., Türkdoğan, D., Altuntaş, Ş. L., Kabil-Kucur, S., **Bulut, O.**, & Kırpınar, G. (2026). SARS-CoV-2 exposure and early communication development: A study of children in Türkiye. *SSRN*. doi:10.2139/ssrn.6908579

13 Abroampah, D., & **Bulut, O.** (2026). Exploring factors contributing to student sociability in Canada: A machine learning approach. *Canadian Journal of School Psychology, 41*(3), 207-233. doi:10.1177/08295735261447107

14 Yildirim-Erbasli, S. N., & **Bulut, O.** (2026). Exploring data augmentation in a low-resource language context: A case study on text generation for reading comprehension in Turkish. *Algorithms, 19*(5), 413. doi:10.3390/a19050413

15 Wei, Y., Sha, L., McWeeny, R., Johnston, K., **Bulut, O.**, Cao, B., Zhang, Y., Baxter, A., Carr, W., Greenshaw, A., Silverstone, P., & Kutcher, S. (2026). Revisiting the knowledge concept of mental health literacy and its relationships with stigma. *Scientific Reports, 16*, Article 19117. doi:10.1038/s41598-026-50222-w

16 **Bulut, O.**, & Zheng, Y. (2026). Editorial: Special issue on big data sources in educational measurement. *Chinese/English Journal of Educational Measurement and Evaluation, 7*(1), Article 1. doi:10.59863/QJYE2767

17 Mazzullo, E., **Bulut, O.**, Jerez, D., Vo, K., Walsh, C., Sitarenios, G., & MacIntosh, A. (2026). Is it worth the effort? A comparison of automatic educational feedback generated by base and fine-tuned LLMs. *Education and Information Technologies, 31*, 5017-5042. doi:10.1007/s10639-026-13988-0

18 **Bulut, O.**, & Yildirim-Erbasli, S. N. (2026). Assessment analytics in digital assessments. *Encyclopedia, 6*(4), Article 81. doi:10.3390/encyclopedia6040081

19 Liu, J. X., Armoush, N., & **Bulut, O.** (2026). Optimizing test completion in time-constrained adaptive tests: A post-hoc simulation study. *Journal of Computerized Adaptive Testing, 13*(1), 1-14. doi:10.7333/2602-1301001

20 Li, W., **Bulut, O.**, Gierl, M. J., & Zhang, S. (2026). Scale abbreviation with supervised machine learning: A comparison of feature selection techniques. *Behavior Research Methods, 58*, Article 55. doi:10.3758/s13428-025-02913-x

21 Jerez, D., Mazzullo, E., & **Bulut, O.** (2026). Exploring slow responses in international large-scale assessments using sequential process analysis. *Computers, 15*(1), Article 64. doi:10.3390/computers15010064

22 **Bulut, O.**, Liu, J. X., & Bulut, H. C. (2026). Disengagement matters: A response-time informed approach to scoring low-stakes assessments. *International Journal of Assessment Tools in Education, 13*(1), 123-144. doi:10.21449/ijate.1721066

23 **Bulut, O.**, & Daniels, L. M. (2025). Explain or diminish: Rethinking generative AI through the lens of

student motivation. *EdArXiv*. doi:10.35542/osf.io/gjeq2_v1

24 **Bulut, O.**, Yildirim-Erbasli, S. N., Ulitzsch, E., & Gorgun, G. (2025). Measuring consumers' financial well-being: Uncovering the causes and consequences of careless and insufficient effort in self-report responses. *Journal of Financial Counseling and Planning, 37*(2), 216-231. doi:10.1891/JFCP-2024-0103

25 Marmolejo-Ramos, F., **Bulut, O.**, Anunciacao, L., Marques, L., Barthakur, A., Kundrat, J., Recka, K., Karakale, O., Correa, J. C., Pinos-Ullauri, L. A., Ospina, R., & Tejada, J. (2025). From human artefact to machine output: Automating the "art" of psychological measurement. *Journal of Psychology and AI, 1*(1), Article 2561692. doi:10.1080/29974100.2025.2561692

26 Wongvorachan, T., **Bulut, O.**, Gorgun, G., & Daniels, L. M. (2025). Evaluating the validity of the student perspectives of teaching survey: A network psychometrics approach. *Trends in Higher Education, 4*(4), 74. doi:10.3390/higheredu4040074

27 Wongvorachan, T., & **Bulut, O.** (2025). Detecting construct-irrelevant variance: A comparison of network psychometrics and traditional psychometric methods using the HEXACO-PI dataset. *Psychology International, 7*(4), 88. doi:10.3390/psycholint7040088

28 Wei, Y., Zhang, Y., Greenshaw, A. J., Cao, B., Bhat, V., **Bulut, O.**, Agyapong, V. I. O., Baxter, A., Li, X.-M., Carr, W., & Ross, D. (2025). Integrating artificial intelligence into school-based mental health approaches and interventions. *World Child & Adolescent Psychiatry, 28*.

29 Cankaya, O., Rohatyn-Martin, N., Buro, K., **Bulut, O.**, & Taylor, K. (2025). Loose parts play encourages spontaneous science, technology, engineering, and mathematics (STEM) behaviours. *Communications Psychology, 3*, Article 183. doi:10.1038/s44271-025-00362-y

30 **Bulut, O.**, & Zheng, Y. (2025). Editorial: Special issue on artificial intelligence and machine learning in educational measurement (Part 3). *Chinese/English Journal of Educational Measurement and Evaluation, 6*(2), Article 1. doi:10.59863/DJYM9770

31 Mazzullo, E., Salma, J., Walsh, C. A., Au, A., Sidani, S., Agrawal, S., **Bulut, O.**, & Guruge, S. (2025). Neighborhood social influences on loneliness in older immigrants in two Canadian provinces. *Canadian Journal on Aging, 44*(4), 449-457. doi:10.1017/S0714980825100068

32 Aydin, B., Kışla, T., Elmas, N. T., & **Bulut, O.** (2025). Automated scoring in the era of artificial intelligence: An empirical study with Turkish essays. *System*, 133, Article 103784. doi:10.1016/j.system.2025.103784.

33 Tan, B., & **Bulut, O.** (2025). Within-person variability in response time as a predictor of test performance: Evidence from TIMSS 2019 problem solving and inquiry tasks. *Journal of Intelligence, 13*(7), 82. doi:10.3390/jintelligence13070082.

34 He, S., **Bulut, O.**, & Cui, Y. (2025). Preschoolers' initial planning time on problem-solving performance across task difficulties: Insights from intelligent tutoring mobile application log data. *Interactive Learning Environments, 34*(3), 1452–1466. doi:10.1080/10494820.2025.2523394

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TECHNICAL REPORTS

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COMPUTER SOFTWARE

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CONFERENCE PRESENTATIONS & POSTERS (LAST 5 YEARS)

1 **Bulut, O.**, Tan, B., Mazzullo, E., & Walsh, C. (2026, October). *The scoring paradox: Multi-agent architectures for unbiased feedback optimization*. Paper presented at the Artificial Intelligence in Measurement and Education Conference (AIME-Con), Pittsburgh, PA.

2 **Bulut, O.**, & Walsh, C. (2026, June). *A multi-agent system for feedback quality assurance: Evaluate, revise, and validate at scale*. Paper presented at the Impactful and Responsible AI Systems for Education Workshop at the Festival of Learning, Seoul, South Korea.

3 **Bulut, O.**, & Walsh, C. (2026, June). *Closing the feedback loop: Automated quality assurance for LLM-generated feedback*. Poster presented at the annual meeting of the International Conference on AI in Education, Seoul, South Korea.

4 **Bulut, O.**, Tan, B., & Yildirim-Erbasli, S. (2026, June). *Does conversation-based assessment elicit higher-level thinking? Evidence from expert and LLM ratings*. Poster presented at the annual meeting of the International Conference on AI in Education, Seoul, South Korea.

5 Yuan, Z., **Bulut, O.**, Geoffre, T., & Wang, Q. (2026, June). *From assessment to human–AI co-creation in language learning: Adaptive, inclusive, and game-based design in the generative AI era*. Workshop presented at the annual meeting of the International Conference on AI in Education, Seoul, South Korea.

6 **Bulut, O.** (2026, June). *Rethinking teaching and assessment in the age of generative AI: Building AI-ready learners and educators*. Paper presented at the Big Thinking Summit 2026: Inflection Point, Edmonton, Alberta, Canada. doi:10.5281/zenodo.20674193

7 Mazzullo, E., & **Bulut, O.** (2026, April). *From proprietary to open-source: A comparative evaluation of LLMs for automatic feedback generation*. Paper presented at the International Conference on Learning Analytics and Knowledge (LAK26), Bergen, Norway.

8 Jerez, D. & **Bulut, O.** (2026, April). *Tracing the paths of slow responders in complex problem solving: A transition network analysis approach*. Paper presented at the International Conference on Learning Analytics and Knowledge (LAK26), Bergen, Norway.

9 Jerez, D., Mazzullo, E., & **Bulut, O.** (2026, April). *Using sequence analysis to explore slow responses in complex problem-solving*. Poster presented at the International Conference on Learning Analytics and Knowledge (LAK26), Bergen, Norway.

10 Kuang, H., **Bulut, O.**, Xiong, J., Botelho, A., & Kern, J. (2026, April). *Seeing how students think: Enhancing assessment through learning analytics of behavioral log and response modeling*. Poster presented at the International Conference on Learning Analytics and Knowledge (LAK26), Bergen, Norway.

11 **Bulut, O.**, Yildirim-Erbasli, S. N., & Tan, B. (2026, April). *Comparing perceived cognitive demand across different item formats*. In M. A. Lindner & D. Zapata Rivera (Chairs), Advancing conversation-based assessment with large language models. Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

12 Sayin, A., & **Bulut, O.** (2026, April). *Impact of essay length on human scoring and chain-of-thought-based automated scoring systems*. Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

13 Mazzullo, E., & **Bulut, O.** (2026, April). *Variability and carelessness in AI-generated psychometric data: A comparison with human responses*. Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

14 Jerez, D., Mazzullo, E., & **Bulut, O.** (2026, April). *Exploring slow responses in interactive problem-solving tasks using sequential process analysis*. Poster presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

15 Liu, J. X., Abroampah, D., & **Bulut, O.** (2026, April). *Exploring response time-based approaches to enhance item selection in time-constrained CATs*. Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

16 Tan, B., Liu, X., & **Bulut, O.** (2026, April). *Investigating student test quitting using response time data*. Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

17 Liu, X., Dong, Y., Tan, B., Cui, Y., & **Bulut, O.** (2026, April). *Detecting intuitive and deliberative thinking modes in PISA 2022 creative thinking assessment*. Poster presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

18 Li, W., & **Bulut, O.** (2026, April). *Which scoring methods should be used for shortened scales?* Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

19 Kuang, H., Xiong, J., **Bulut, O.**, & Kern, J. (2026, April). *A bi-factor model for latent trait estimation integrating responses and process data*. Paper presented at the annual meeting of the National Council on Measurement in Education, Los Angeles, CA.

20 Gunduz, A., Gierl, M. J., & **Bulut, O.** (2025, October). *Enhancing essay scoring with GPT-2 using back translation techniques*. Paper presented at the Artificial Intelligence in Measurement and Education Conference (AIME-Con), Pittsburgh, PA.

21 Gorgun, G., **Bulut, O.**, & Lindner, M. A. (2025, August). *Response behavior is a good proxy for difficulty estimation in adaptive learning environments*. Paper presented at the European Association for Research on Learning and Instruction (EARLI) Conference, Graz, Austria.

22 **Bulut, O.**, Liu, J. X., & Demir, H. (2025, July). *Deep reinforcement learning for engagement-aware question selection in adaptive assessment systems*. Poster presented at the International Conference on Artificial Intelligence in Education, Palermo, Italy.

23 Ulitzsch, E., & **Bulut, O.** (2025, June). *Detecting careless responding in the PIRLS 2021 student questionnaire: An IRT mixture modeling approach*. Paper presented at the IEA International Research Conference (IEA IRC), Rome, Italy.

24 Vo, K., & **Bulut, O.** (2025, June). *Understanding undergraduate strategies in ChatGPT-assisted academic writing for the social sciences: A grounded theory and observational study*. Poster presented at the annual meeting of the Canadian Society for the Study of Higher Education (CSSHE), Toronto, ON, Canada.

25 Rinaldi, C. M., Beeby, A. M., **Bulut, O.**, Gokiert, R. J., & Howe, N. (2025, June). *Parenting stress in early childhood: A comparison of mothers' and fathers' self-reported parent-child dysfunction from preschool to grade one*. Poster presented at the annual meeting of the Alberta Social Psychology Symposium, Calgary, AB.

26 Morris, H., Cottrell-McDermott, C., Taylor, M., Hyshka, E., **Bulut, O.**, & Salvalaggio, G. (2025, June). *Assessing the impact of the COVID-19 pandemic on people who use drugs in Edmonton, Canada: Implications for community health nursing practice*. Poster presented at the International Council of Nursing Congress, Helsinki, Finland.

27 Smith-Ackerl, S., Mack, E., Sevigny, P., **Bulut, O.**, & Wallin, J. (2025, June). *Fictional stories with real*

consequences? A study on youth, media, supportive adults, and suicide. Paper presented at the annual general meeting and national convention of the Canadian Psychological Association, St. John's, NL.

28 Ahmed, N., Akdemir, K., Kim, M., Wong, S., & **Bulut, O.** (2025, June). *The complexity of elementary students' perspective-taking and decision-making on environmental issues*. Paper presented at the CSSE Conference, George Brown College, Toronto, ON, Canada.

29 Dufresne, D., Jerez, D., Makowecki, E., Wuest, V., Rabbani, K., Dann, A., Fenwick, D., Martin, R., McMorris, C., Milne, D., Mullen-Wawryk, L., Murphy, L., Ray, A., Sanders, J., Seale, R., Tremblay, M., Pei, J., **Bulut, O.**, & Poth, C. (2025, May). *Prioritizing collaboration in survey development: Lessons from a community-based approach*. 2025 UAlberta Graduate Conference in Education, Edmonton, Alberta.

30 Hall, M., **Bulut, O.**, Dick, B., Pham, A., & Jones, A. (2025, May). *Burnout in Canadian physiotherapists during COVID-19 - A national survey*. Poster presented at the World Physiotherapy Congress, Tokyo, Japan.

31 Mazzullo, E., & **Bulut, O.** (2025, April). *Fine-tuning GPT-3.5-Turbo for automatic feedback generation*. Paper presented at the 40th ACM/SIGAPP Symposium on Applied Computing (ACM SAC 2025): Artificial Intelligence for Education, Sicily, Italy.

32 **Bulut, O.** (2025, April). *AI in educational measurement: Addressing challenges and ensuring responsible use*. In J. Whitmer (Chair), Generative AI in education: Applied ethics & evaluating model performance. Paper presented at the annual meeting of the National Council on Measurement in Education, Denver, CO.

33 Liu, J. X., & **Bulut, O.** (2025, April). *Sequential modeling of responses and response times in a CAT with LSTM*. In H. Kuang (Chair), Fuse diverse process data in educational measurement. Paper presented at the annual meeting of the National Council on Measurement in Education, Denver, CO.

34 Mazzullo, E., & **Bulut, O.** (2025, April). *Fine-tuning GPT and Llama for automatic feedback generation: A comparative study*. In G. Gorgun (Chair), Leveraging large-language models for operational large-scale assessments: Opportunities and challenges. Paper presented at the annual meeting of the National Council on Measurement in Education, Denver, CO.

35 Gorgun, G., & **Bulut, O.** (2025, April). *Predicting the item characteristics of scenario-based assessment items*. In G. Gorgun (Chair), Leveraging large-language models for operational large-scale assessments: Opportunities and challenges. Paper presented at the annual meeting of the National Council on Measurement in Education, Denver, CO.

36 Armoush, N., Liu, J. X., & **Bulut, O.** (2025, April). *Optimizing test completion in strictly-timed CATs: A post-hoc simulation*. Paper presented at the annual meeting of the National Council on Measurement in Education, Denver, CO.

37 He., S., **Bulut, O.**, & Cui, Y. (2025, April). *Preschooler's planning time on problem-solving performances across different task difficulties: Insights from mobile log data*. Paper presented at the annual meeting of the American Educational Research Association, Denver, CO.

38 Morris, H., Cottrell-McDermott, C., Piggott, B., Hyshka, E., Taylor, M., **Bulut, O.**, Connolly, D., Staines, A., Issa, T., Gehring, N., Collins, Z., Raza, A. & Salvalaggio, G. (2025, January). *Understanding the short- and long-term impacts of the COVID-19 pandemic on people who use substances in Edmonton's inner city: Findings of the 2023 ICHWP cross-sectional survey*. City Wide Nursing Rounds, Faculty of Nursing, University of Alberta.

39 Mazzullo, E., & **Bulut, O.** (2024, December). Automated feedback generation for open-ended questions: Insights from fine-tuned LLMs. Paper presented at the NeurIPS 2024 Workshop on Large Foundation Models for Educational Assessment, Vancouver, BC.

40 Connoly, D., Morris, H., Piggott, B., Cottrell-McDermott, C., Hyshka, E., Taylor, M., **Bulut, O.**, Issa, T., & Salvalaggio, G. (2024, November). Co-occurring issues facing patients who use unregulated drugs: Insights from a survey in Edmonton, Canada. Poster presented at the annual meeting of the North American Primary Care Research Group, Quebec City, QC.

41 Svetina Valdivia, D., **Bulut, O.**, Withrow, A., & Walsh, P. (2024, August). Impact of reversals in categories on IRT model fit and parameter recovery. Poster presented at the annual meeting of the American Psychological Association, Seattle, WA.

42 Hunt, T. E., Waldron, M. B. K., Tejada, J., Kotera, Y., Sari, M. H., Asanjarani, F., Ashraf, F., Muwonge, C. M., Popa,I-L., Bagdasar, O., Gabriel, F., Kojima, Y., Vione, K. C., Sheffield, D., **Bulut, O.**, & Marmolejo-Ramos, F. (2024, July). Numbers without borders: A comparative study of mathematics anxiety and its predictive factors. Paper presented at the 4th International Conference Opportunities and Challenges of Building Mathematical Resilience, Coventry, UK.

43 **Bulut, O.**, Clelland, A.C., & Wongvorachan, T. (2024, June). *Identifying the predictors of mathematics performance and math anxiety in Canada: An educational data mining approach*. Paper presented at the annual meeting of the Canadian Society for the Study of Education, Montreal, QC.

44 **Bulut, O.**, Shin, J., & Pinto Junior, W. N. (2024, April). *Towards fair learning analytics: An application of the Seldonian algorithm*. In Q. He (Chair), Incorporating multiple data sources to explore group strategy variations. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

45 **Bulut, O.**, Gorney, K., & Yildirim-Erbasli, S. N. (2024, April). *Joint modeling of ability and test-taking effort with multidimensional nested logit model*. In B. Arslan (Chair), Unpacking test-taker (dis)engagement: Innovative approaches to enhance validity. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

46 Gorgun, G., & **Bulut, O.** (2024, April). *Using large language models for evaluating item quality in large-scale assessments*. In S. Dai (Chair), Innovations and practices of AI and language models in large-scale assessments. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

47 Firoozi, T., **Bulut, O.**, & Gorgun, G. (2024, April). *Text augmentation for enhancing the accuracy of automated essay scoring*. In H. Jiao & X. Zhai (Chairs), Tackling challenges in automated scoring: Data augmentation, multilingualism, and feedback mechanisms. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

48 Liu, X., **Bulut, O.**, Gao, Y., & Cui, Y. (2024, April). *Uncovering test-takers' problem-solving patterns through Hidden Markov models and network analysis*. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

49 Clelland, A.C., & **Bulut, O.** (2024, April). *Using network psychometrics to identify wording effects in a socioemotional skills survey*. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

50 Tan, B., Armoush, N., **Bulut, O.**, Mazzullo, E., & Gierl, M. (2024, April). *Leveraging pre-trained large language models for automatic item generation: A scoping review*. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

51 Mazzullo, E., **Bulut, O.**, Tan, B., & Wongvorachan, T. (2024, April). *How can large language models revolutionize learning analytics?* Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

52 Wongvorachan, T., **Bulut, O.**, & Gorgun, G. (2024, April). *Evaluating dimensionality in non-cognitive measures: Exploratory vs. hierarchical exploratory graph analysis.* Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

53 Clelland, A.C., & **Bulut, O.** (2024, April). *A network psychometrics approach to explore Canadian students' collaborative skills.* Poster presented at the annual meeting of the American Educational Research Association, Philadelphia, PA.

54 Jones, A., **Bulut, O.**, Dick, B., Pham, A., & Hall, M. (2024, April). *Burnout amongst Canadian physiotherapists during the COVID-19 pandemic: A national survey*. Paper presented at the Canadian Physiotherapy Association Congress, Vancouver, BC.

55 Vo, K., & **Bulut, O.** (2024, March). *AI-assisted talk: A narrative review on the new social and conversational landscape*. Poster presented at the Spring Symposium Series of the Association for the Advancement of Artificial Intelligence, Stanford, CA.

56 Vo, K., & **Bulut, O.** (2024, March). *Social smarts with tech sparks: Harnessing LLMs for youth socio-emotional growth*. Poster presented at the Spring Symposium Series of the Association for the Advancement of

Artificial Intelligence, Stanford, CA.

57 Ferede, S., Akdemir, K., Ahmed, N., Jin, Q., MacKay, J., Wong, S., **Bulut, O.**, & Kim, M. (2024, March). *Knowledge, perspectives, and decision making on climate change in science classrooms*. Poster presented at the Celebrating Socially Transformative Research and Teaching Event at the University of Alberta, Edmonton, AB.

58 Connolly, D., Morris, H., Staines, A., Piggott, B., Cottrell-McDermott, C., Hyshka, E., Taylor, M., **Bulut, O.**, Issa, T., Collins, Z., & Salvalaggio, G. (2024, March). *Adjusting harm reduction services to meet the needs of people who consume drugs through inhalation in central Edmonton*. Poster presented at the Alberta College of Family Medicine's Family Medicine Summit, Banff, AB.

59 Gorgun, G., & **Bulut, O.** (2024, February). *Current evaluation methods are a bottleneck in automatic question generation*. Poster presented at the Workshop on AI for Education: Bridging Innovation and Responsibility at the AAAI Annual Conference on AI, Vancouver, BC.

60 Yürekli, H., & **Bulut, O.** (2023, November). *Unveiling hidden patterns in students' reading achievement through data mining*. Paper presented at the 7th International Researchers, Statisticians, and Young Statisticians Congress, Yildiz Technical University, Istanbul, Türkiye.

61 **Bulut, O.** (2023, October). *Large language models in educational assessment*. Panel talk given at the Empowering Learners for the Age of AI Online Conference.

62 Ivan, R., MacIntosh, A., **Bulut, O.**, Shin, J., Musolino, G. M., Traywick, L. S., Salamh, P. A., Condo, M. P., & Herringer, C. N. (2023, October). *Ethical AI in health professions education: Co-creating an ethical framework to ensure responsible AI use*. Paper presented at the annual Physical Therapy Education Leadership Conference, Philadelphia, PA.

63 Yildirim-Erbasli, S. N., & **Bulut, O.** (2023, July). *Innovating assessment with conversational agents: A technology-enhanced approach to formative assessments*. Paper presented at the 23rd IEEE International Conference on Advanced Learning Technologies, Orem, UT.

64 **Bulut, O.**, & Gorgun, G. (2023, June). *Utilizing response time for scoring the TIMSS 2019 problem solving and inquiry tasks*. Paper presented at the IEA International Research Conference (IEA IRC), Dublin, Ireland.

65 **Bulut, O.**, Gorgun, G., Tan, B., Wongvorachan, T., Yildirim-Erbasli, S., & Clelland, A. (2023, June). *Self-regulated learning through online formative assessments: The effects of assessment frequency and participation on student performance*. Paper presented at the annual meeting of the International Society of the Learning Sciences, Montréal, QC.

66 Yildirim-Erbasli, S. N., Demmans Epp, C., **Bulut, O.**, & Cui, Y. (2023, June). *Design and evaluation of a conversational agent for formative assessment in higher education*. Paper presented at the annual meeting of the International Society of the Learning Sciences, Montréal, QC.

67 Rinaldi, C., Keys, K., **Bulut, O.**, Gokiert, R., & Howe, N. (2023, June). *From puzzles to put away: Examining affect and responsiveness in parent-toddler interactions across different task contexts*. Poster presented at the annual conference of the Jean Piaget Society, Madrid, Spain.

68 Steele, B. J., Sidney, B., Menard, K., **Bulut, O.**, Lowe, S., Pabayo, R., & Yamamoto, S. (2023, June). *Development and initial validation of the ACCLIMATES surveillance tool: Assessing climate change-related chronic disease vulnerability in Alberta, Canada*. Poster presented at the annual meeting of the Society for Epidemiologic Research, Portland, OR, USA.

69 **Bulut, O.**, MacIntosh, A., & Walsh, C. (2023, April). *Utilizing NLP techniques for collecting validity evidence for a situational judgement test*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

70 **Bulut, O.**, Yildirim-Erbasli, S. N., He, S., Tan, B., & Gao, Y. (2023, April). *Adaptive item recommendation using process data and examinee background characteristics*. In Q. He & O. Bulut (Chairs), Empowering process data for data-informed decision-making in measurement. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

71 Gorgun, G., **Bulut, O.**, & Wongvorachan, T. (2023, April). *Detecting rapid guessing: In search of the optimal response time threshold*. In E. Ulitzsch & J. Rios (Chairs), Methodological advances in detecting and

accounting for noneffortful responding. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

72 Yildirim-Erbasli, S. N., **Bulut, O.**, & Cui, Y. (2023, April). *Conversation-based assessments: Enhancing students' experiences of formative assessment via human-like interaction*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

73 Bulut, H. C., & **Bulut, O.** (2023, April). *Exploring added-value of response times for low-stakes assessments with nested logit models*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

74 Gorgun, G., Wongvorachan, T., Tan, B., & **Bulut, O.** (2023, April). *Predicting item difficulty from students' response behaviors*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

75 Goodrich, S., & **Bulut, O.** (2023, April). *Dynamic, on-the-fly, and hybrid multistage testing with the bi-factor model*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

76 Kilmen, S., **Bulut, O.**, & Shin, J. (2023, April). *Creating short forms of psychological scales: Semantic similarity matters*. Poster presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

77 Yildirim-Erbasli, S. N., Gorgun, G., & **Bulut, O.** (2023, April). *Automatic item paraphrasing to avoid the repetitive use of the same items*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

78 Sayin, A., & **Bulut, O.** (2023, April). *Estimating item difficulty: What variables do subject matter experts use?* Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

79 He, S., Wongvorachan, T., **Bulut, O.**, & Lai, K.-W. (2023, March). *Identifying early warning indicators for high school dropouts*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

80 Tan, B., **Bulut, O.**, Gorgun, G., & Wongvorachan, T. (2023, March). *Mining textual features of questions to predict item parameters*. Poster presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

81 Wongvorachan, T., He, S., Lai, K. W., & **Bulut, O.** (2023, March). *A comparison of sampling methods for imbalanced classifications in educational datasets*. Poster presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

82 Wongvorachan, T., **Bulut, O.**, Gorgun, G., & Tan, B. (2023, March). *Random search algorithm to identify response time thresholds for rapid guessing*. Paper presented at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

83 Rinaldi, C., Martinovich, V., Durber, C., & **Bulut, O.** (2023, March). *Mothers' and fathers' parenting and involvement reports from preschool to first grade*. Poster presented at the Society for Research in Child Development Biennial Meeting, Salt Lake City, Utah.

84 MacIntosh, A., Walsh, C., & **Bulut, O.** (2022, August). *Using natural language processing to understand the multidimensionality of professionalism through a constructed-response situational judgement test*. Paper presented at the Ottawa 2022 conference.

85 Wongvorachan, T., Poth, C., **Bulut, O.**, & Otto, S. (2022, August). *Leveraging open data sources in mixed methods research*. Paper presented at the biennial global conference of the Mixed Methods International Research Association (MMIRA).

86 Shin, J., **Bulut, O.**, & Pinto, W. N. (2022, July). *E-learning preparedness: A key consideration to promote fair learning analytics development in higher education*. Paper presented at the Educational Data Mining Conference, Durham, UK.

87 **Bulut, O.**, MacIntosh, A., & Walsh, C. (2022, July). *Leveraging natural language processing for quality assurance of a situational judgement test*. Paper presented at the annual meeting of the Artificial

Intelligence in Education, Durham, UK.

88 Gorgun, G., & **Bulut, O.** (2022, July). *Considering disengaged student behavior in Bayesian and deep knowledge tracing*. Poster presented at the annual meeting of the Artificial Intelligence in Education, Durham, UK.

89 Ulitzsch, E., Yildirim-Erbasli, S. N., Gorgun, G. & **Bulut, O.** (2022, July). *An explanatory mixture IRT model for careless and insufficient effort responding in self-report measures*. Paper presented at the annual meeting of the Psychometric Society, Bologna, Italy.

90 McCabe, E., Gross, D. P., & **Bulut, O.** (2022, June). *Improving the efficiency of patient-reported outcome measurement: Procedures for developing a computerized adaptive test to assess physical functioning*. Paper presented at the annual conference of the Canadian Association for Health Services and Policy Research.

91 **Bulut, O.**, & Wongvorachan, T. (2022, May). *Feedback generation through artificial intelligence*. Paper presented at the annual meeting of Open/Technology in Education, Society, and Scholarship Association (OTESSA).

92 Hachem, M., Gorgun, G., Chu, M-W., & **Bulut, O.** (2022, April). *Social and emotional variables as predictors of students' perceived cognitive competence and academic performance*. Paper presented at the uLead Summit of Educational Leadership, Banff, AB.

93 Deforge, K., Seitz, P., **Bulut, O.**, Chu, M-W., Gorgun, G., & Hachem, M. (2022, April). *Giving students a voice: A measure of students' socio-emotional wellbeing to optimize learning for all students*. Paper presented at the uLead Summit of Educational Leadership, Banff, AB.

94 **Bulut, O.**, & Yildirim-Erbasli, S. N. (2022, April). *Detecting careless responses with the rapid guessing-slow responding method*. Paper presented at the annual meeting of the National Council on Measurement in Education, San Diego, CA.

95 Bulut, H. C., & **Bulut, O.** (2022, April). *A network psychometrics approach for detecting item wording effects across subgroups*. Paper presented at the annual meeting of the National Council on Measurement in Education, San Diego, CA.

96 Gorgun, G., & **Bulut, O.** (2022, April). *A smarter CAT: Integrating engagement into the CAT item selection algorithm*. Paper presented at the annual meeting of the National Council on Measurement in Education, San Diego, CA.

97 Firoozi, T., Abadi, A. N., Demmans Epp, C., **Bulut, O.**, & Barbosa, D. (2022, April). *The effect of word vector representation and linguistic features on the accuracy of automated essay scoring systems using neural networks*. Paper presented at the annual meeting of the National Council on Measurement in Education, San Diego, CA.

98 Yildirim-Erbasli, S. N., Gorgun, G., Gao, Y., & **Bulut, O.** (2022, April). *Sequential pattern mining to understand time management strategies in online courses*. Poster presented at the annual meeting of the National Council on Measurement in Education, San Diego, CA.

99 Ulitzsch, E., Yildirim-Erbasli, S. N., Gorgun, G. & **Bulut, O.** (2022, April). *An explanatory mixture IRT model for careless responding in survey data*. Paper presented at the annual meeting of the National Council on Measurement in Education, San Diego, CA.

100 Tobias, A. D., Cutumisu, M., & **Bulut, O.** (2022, April). *Estimating classification accuracy and decision consistency with classdecision*. Paper presented at the annual meeting of the American Educational Research Association, San Diego, CA.

101 **Bulut, O.**, Yildirim-Erbasli, S. N., Gorgun, G., Gao, Y., Wongvorachan, T., Lai, K. W., & Shin, J. (2022, March). *Leveraging online formative assessments to enhance predictive learning analytics models*. Paper presented at the International Conference on Learning Analytics and Knowledge (LAK2022).

102 Walsh, C., MacIntosh, A., **Bulut, O.**, & Shin, J. (2022, March). *What are we measuring?: A topic modeling framework to map professionalism aspects to responses in a situational judgment test*. Poster presented at the International Conference on Learning Analytics and Knowledge (LAK2022).

103 Wongvorachan, T., Clelland, A., Gorgun, G., **Bulut, O.** (2022, April). *Identifying the predictors of mathematics anxiety and performance in Canada: An educational data mining approach*. Poster presented at the 14th Annual Graduate Student Research Showcase: Exploring Horizons (University of Alberta),

Edmonton, AB, Canada.

104 Nielsen, C., Tilstra, M., **Bulut, O.**, Vargas, A. O., & Yamamoto, S. (2022, April). *Community climate change health tool for populations experiencing higher vulnerability*. Paper presented at the Geodesign Summit.

105 Wongvorachan, T., Lai, K. W., **Bulut, O.**, Tsai, Y.S., Chen, G. (2022, February). *The application of artificial intelligence in educational feedback.* Poster presented at Reverse Expo 2022 (University of Alberta), Edmonton, AB, Canada.

106 Yildirim-Erbasli, S. N., & **Bulut, O.** (2022, January). *A conversational agent to measure student learning: Conversation-based assessment*. Poster presented at the Advancing Assessment and Evaluation Conference (AAEC).

107 Wongvorachan, T., Lai, K.W., **Bulut, O.**, Tsai, Y. S. (2022, January). *Feedback literacy framework in the e-assessment context*. Poster presented at the Advancing Assessment and Evaluation Conference (AAEC).

108 Gorgun, G. & **Bulut, O.** (2022, January). *Customizing item selection with engagement in computerized-adaptive tests*. Poster presented at the Advancing Assessment and Evaluation Conference (AAEC).

109 **Bulut, O.**, & Yildirim-Erbasli, S. (2021, November). Real-time assessment and feedback for real-time learning. Paper presented at the Beyond Multiple Choice Conference.

110 Yildirim-Erbasli, S. N., & **Bulut, O.** (2021, October). *Automatic story generation for reading comprehension*. Poster presented at the annual meeting of the National Council on Measurement in Education (NCME) Special Conference on Classroom Assessment.

111 Tilstra, M., Nielsen, C., Tiwari, I., Azizali, S., Jones, A., Quemerais, B., Vargas, A. O., **Bulut, O.**, Salma, J., Yamamoto, S. (2021, August). *Exploring community-level health impacts of extreme temperatures and air pollution in older adult and immigrant populations living in Edmonton, AB*. Poster presented at the annual conference of the International Society for Environmental Epidemiology.

112 Nielsen, C. C., Tilstra, M., **Bulut, O.**, Vargas, A. R. O, & Yamamoto, S. (2021, August). *The intersection of climate-related health outcomes and urban environmental characteristics using spatiotemporal data science*. Paper presented at the annual conference of the International Society for Environmental Epidemiology.

113 Nielsen, C. C., Tilstra, M., **Bulut, O.**, Vargas, A. R. O, & Yamamoto, S. (2021, June). *Spatiotemporal hot spot associations of climate-related health and urban environment*. Paper presented at the annual conference of the Canadian Association of Geographers.

114 Tilstra, M., Nielsen, C., Tiwari, I., Jones, A., Quemerais, B., Vargas, A. O., **Bulut, O.**, Salma, J., & Yamamoto, S. (2021, June). *Characterizing local climate change vulnerability: Impacts of climatic and air pollution exposures on older adults and immigrants in Edmonton, Alberta*. Paper presented at the annual meeting of the Canadian Society for Epidemiology and Biostatistics.

115 **Bulut, O.**, Cormier, D. C., & Shin, J. (2021, June). *The end of one-size-fits-all testing: Personalizing test schedules with recommender systems*. Paper presented at the annual meeting of the National Council on Measurement in Education.

116 Gorgun, G., Yildirim Erbasli, S. N.,& **Bulut, O.** (2021, June). *An application of explanatory IRT to predict factors contributing to rapid guessing*. Paper presented at the annual meeting of the National Council on Measurement in Education.

117 Yildirim Erbasli, S. N., Gorgun, G., & **Bulut, O.** (2021, June). *Detecting careless responses: A new method utilizing response time and accuracy*. Paper presented at the annual meeting of the National Council on Measurement in Education.

118 Gao, Y., **Bulut, O.**, & Cui, Y. (2021, June). *Diagnosing adult information problem solving strategy by mining clickstream data*. Paper presented at the annual meeting of the National Council on Measurement in Education.

119 Gao, Y., Xiaoming, Z., **Bulut, O.**, & Cui, Y. (2021, June). *Understanding problem-solving styles in technology-rich environments by log data analysis*. Poster presented at the annual meeting of the National Council on Measurement in Education.

120 Bulut, H. C., Arikan, S., & **Bulut, O.** (2021, June). *Student and item characteristics in online reading*

comprehension: Polytomous explanatory IRT modeling. Paper presented at the annual meeting of the National Council on Measurement in Education.

121 Firoozi, T., Shin, J., **Bulut, O.**, & Gierl, M. J. (2021, June). *The effect of option similarity on the item difficulty of a reading comprehension test: An application of word embedding techniques*. Paper presented at the annual meeting of the National Council on Measurement in Education.

122 Hachem, M., Gorgun, G., Chu, M.-W., & **Bulut, O.** (2021, June). *Social and emotional variables as predictors of students' cognitive competence and academic performance*. Paper presented at the annual meeting of the Canadian Society for the Study of Education.

123 Philippot, C. M., Antoniuk, A., Cormier, D. C., Yildirim-Erbasli, S. N., & **Bulut, O.** (2021, June). *Opening the vault: Using a projective measure to access the financial attitudes of young people*. Poster presented at the annual national conference of the Canadian Psychological Association, Montreal, QC.

124 Antoniuk, A., Philippot, C. M., Cormier, D. C., Yildirim-Erbasli, S. N., & **Bulut, O.** (2021, June). *Measuring financial literacy: Investing in positive financial futures*. Poster presented at the annual national conference of the Canadian Psychological Association, Montreal, QC.

FACULTY & UNIVERSITY SERVICE

- Member; Provost's Taskforce on AI and the Learning Environment — 2025 to Now
- Instructor; FGSR's pedagogical training session on assessment techniques — 2023 to Now
- Member; Committee on the Learning Environment's SPOT Survey Taskforce — 2022 to 2024
- Reviewer; Research Ethics Board 2 at the University of Alberta — 2021 to 2025
- Chair; The HT Coutts Library Advisory Committee — 2019 to Now
- Member; Dean's Pool for Department Advisory Selection Reps — 2020 to 2025
- Program Coordinator; Measurement, Evaluation, and Data Science — 2020 to 2022
- Member; Academic Appeals Committee at the Faculty of Education — 2019 to 2021
- Member; The SSHRC Doctoral Awards Evaluation Committee — 2018 to 2020
- Member; Alberta Advisory Committee on Education Studies — 2017 to 2019
- Member; School of Library & Information Studies Council — 2017 to 2019
- Member; Faculty of Education Teaching Awards Committee — 2015 to 2017
- Member; Interdisciplinary Committee on Applied Graduate Statistics — 2015 to Now

EDITORIAL SERVICE

- Associate Editor, Intelligence
- Associate Editor, Chinese/English Journal of Educational Measurement and Evaluation
- Section Editor-in-Chief; Contributions to the Measurement of Intelligence, Journal of Intelligence
- Special Issue Editor; Frontiers in Education (2023)
- Special Issue Editor; Journal of Intelligence (2023)
- Special Issue Editor; Journal of Applied Testing Technology (2023)
- Associate Editor; Journal of Measurement and Evaluation in Education and Psychology
- Associate Editor; Chinese/English Journal of Educational Measurement and Evaluation
- Editorial Board Member; International Journal of Assessment Tools in Education
- Editorial Board Member; Canadian Journal of Educational Psychology
- Editorial Board Member; Journal of Intelligence
- Editorial Board Member; Psych
- Consulting Editor; Journal of Educational Psychology
- Ad Hoc Journal Reviewer; Educational and Psychological Measurement
- Ad Hoc Journal Reviewer; Applied Psychological Measurement
- Ad Hoc Journal Reviewer; Journal of Educational Measurement
- Ad Hoc Journal Reviewer; Educational Measurement: Issues and Practice
- Ad Hoc Journal Reviewer; International Journal of Testing

OTHER PROFESSIONAL SERVICE

- NAEP Advisory Committee member — 2025 to Now
- Technical advisor for the New Mexico Department of Education — 2025 to Now
- Peer reviewer for the US Department of Education's Peer Assessment Process — 2022 to Now
- Peer reviewer for NRS Assessments at American Institutes for Research — 2020 to Now
- External reviewer for the Early Researcher Awards Program in Ontario — 2022
- External reviewer for the CANSSI Collaborative Research Team Program — 2022
- External reviewer for a Tenure Promotion Package at the University of Tennessee — 2022
- External reviewer for a Tenure Promotion Package at the University of Saskatchewan — 2020
- Organizer; CanAm Online Symposium Series — 2015 to 2024
- Member; The Evaluation Advisory Committee for Youth ART Project — 2015 to 2018

PROFESSIONAL MEMBERSHIP

- National Council on Measurement in Education (NCME), Member — 2009 to Now
- NCME Publications Committee, Member — 2021 to 2024
- American Educational Research Association (AERA), Member — 2009 to Now
- The International Committee for AERA Division D, Member — 2017 to 2020
- Association for Computing Machinery (ACM), Member — 2021 to Now
- Society for Learning Analytics Research (SoLAR), Member — 2022 to Now